

<https://github.com/SophCollenburg/ser321-fall26C-scollenb.git>

Task 1- I am using the Windows Command Prompt

```
mkdir cli_assignment
cd cli_assignment
type nul> stuff.txt
echo WHAT THE HELI? > stuff.txt
( echo I love cats! echo Where do I go for CAVA? echo ao;ehigna;oshaw'eofh a;u hfaiuf 'h a'hoh) >>
stuff.txt
echo Um.. >> stuff.txt
echo this is weird :( >> stuff.txt
find /c /v "" stuff.txt // this only displayed the lines
powershell (Get-Content stuff.txt | Measure-Object -Word).Words // I had to enter windows
powershell to get word count
mkdir draft
move stuff.txt draft
type nul> secret.txt
attrib +h secret.txt
cd ..
mkdir final
xcopy draft final /E /I
move draft draft.remove
move draft final
dir /a /s
attrib /s
```

```
type nul> logs.txt
(echo Was my freedom a borrow
More? echo screamiing at hte tp of my lungs
More? echo all of my sins keep wiehgin my down
More? echo the end has come
More? echo that is the place
More? echo how do I stop
More? echo whty do i live
More? echo how many until this stops? ) >> logs.txt
powershell -Command "Get-Content stuff.txt -TotalCount 5" // this was executed in powershell
(for /f "tokens=* delims=" %a in ('type logs.txt ^| findstr /n "" ^| findstr "[1-5]:") do @echo %a) >
logs_top_5.txt
powershell -Command "Get-Content stuff.txt -Tail 5" // this was executed in powershell
for /f %A in ('find /c /v "" ^< logs.txt') do set lines=%A
```

```
set /a start=%lines%-4
(for /f "tokens=1* delims=" %A in ('findstr /n "^" logs.txt') do @if %A geq %start% echo %A: %B) >
logs_bottom_5.txt
copy logs_top_5.txt + logs_bottom_5.txt logs_snapshot.txt
type logs.txt | more
(for /f "skip=1 tokens=1 delims=" %A in (marks.csv) do @echo %A) > student_names.txt
(for /f "skip=1 tokens=4 delims=" %A in (marks.csv) do @echo %A) | sort > subject3_sorted.txt
for /f "skip=1 tokens=3 delims=" %A in (marks.csv) do @echo %A > temp.txt
powershell -Command "(Get-Content temp_marks.txt | Measure-Object -Sum | ForEach-Object { $_.Sum
/ (Get-Content temp_marks.txt).Count })" > subject2_average.txt
del temp.txt
copy subject2_average.txt done.txt
move done.txt cli_assignment
cd cli_assignment
move done.txt average.txt
```

Task 2

2.2 -

1.

- a) JavaGradle, the Fraction.java class

```

[ec2-user@ip-172-31-31-40 java]$ javac -d . -g Fraction.java
[ec2-user@ip-172-31-31-40 java]$ jdb -classpath . Fraction
Initializing jdb ...
> stop in Fraction.main
Deferring breakpoint Fraction.main.
It will be set after the class is loaded.
> stop in Fraction.<init>
Deferring breakpoint Fraction.<init>.
It will be set after the class is loaded.
> stop in Fraction.setNumerator
Deferring breakpoint Fraction.setNumerator.
It will be set after the class is loaded.
> run
run Fraction
Set uncaught java.lang.Throwable
Set deferred uncaught java.lang.Throwable
>
VM Started: Set deferred breakpoint Fraction.setNumerator
Set deferred breakpoint Fraction.<init>
Set deferred breakpoint Fraction.main

Breakpoint hit: "thread=main", Fraction.main(), line=44 bci=0
44      Fraction frac = new Fraction();

main[1] cont
>
Breakpoint hit: "thread=main", Fraction.<init>(), line=16 bci=0
16      public Fraction(){

main[1] cont
>
Breakpoint hit: "thread=main", Fraction.setNumerator(), line=25 bci=0
25      numerator = n;

main[1] cont
> The fraction is: 1/3

The application exited
[ec2-user@ip-172-31-31-40 java]$ |

```

- b)
- c) The example computes a fraction based off predetermined values to return a result

2.

- a) Threads/Account

```

[ec2-user@ip-172-31-31-40 network]$ cd ..
[ec2-user@ip-172-31-31-40 ser321examples]$ cd Threads
[ec2-user@ip-172-31-31-40 Threads]$ cd Account
[ec2-user@ip-172-31-31-40 Account]$ cd src/main/java
[ec2-user@ip-172-31-31-40 java]$ javac Transaction.java
[ec2-user@ip-172-31-31-40 java]$ ls
Account.class  Transaction.class  Transaction.java
[ec2-user@ip-172-31-31-40 java]$ java Transaction
Expected Arguments: <transactions(int)> <sleep(int)> <deposits(int)>
[ec2-user@ip-172-31-31-40 java]$ java Transaction 4 5 9
Transaction started #1
Transaction started #2
Transaction started #3
Transaction started #4
Balance is 360
[ec2-user@ip-172-31-31-40 java]$ |

```

- b)
- c) This example processed a set of transactions based on user provided inputs.

3.

- a) Threads/Deadlock

```
Transaction started #4
Balance is 360
[ec2-user@ip-172-31-31-40 java]$ cd ..
[ec2-user@ip-172-31-31-40 main]$ cd ..
[ec2-user@ip-172-31-31-40 src]$ cd ..
[ec2-user@ip-172-31-31-40 Account]$ cd ..
[ec2-user@ip-172-31-31-40 Threads]$ cd Deadlock
[ec2-user@ip-172-31-31-40 Deadlock]$ cd src/main/java
[ec2-user@ip-172-31-31-40 java]$ javac Deadlock.java
[ec2-user@ip-172-31-31-40 java]$ ls
'Deadlock$1.class' 'Deadlock$2.class' 'Deadlock$Friend.class'  Deadlock.class  Deadlock.java
[ec2-user@ip-172-31-31-40 java]$ java Deadlock
Alphonse: Gaston  has bowed to me!
Gaston: waiting to bow back
Gaston: Alphonse  has bowed to me!
Alphonse: waiting to bow back
```

b)

c) This example shows what it means to deadlock within a terminal, forcing the user to close it.

2.3

```
ec2-user@ip-172-31-31-40:~/t X + v
runApp - Tasks which runs Multiply with default parameters
runAppAgain - Tasks which runs Multiply with default parameters or given values

Verification tasks
-----
check - Runs all checks.
test - Runs the test suite.

Rules
-----
Pattern: clean<TaskName>: Cleans the output files of a task.
Pattern: build<ConfigurationName>: Assembles the artifacts of a configuration.

To see all tasks and more detail, run gradle tasks --all

To see more detail about a task, run gradle help --task <task>

[Incubating] Problems report is available at: file:///home/ec2-user/ser321-fall26C-scollenb/Assignment
1/build/reports/problems/problems-report.html

Deprecated Gradle features were used in this build, making it incompatible with Gradle 9.0.

You can use '--warning-mode all' to show the individual deprecation warnings and determine if they com
e from your own scripts or plugins.

For more on this, please refer to https://docs.gradle.org/8.13/userguide/command_line_interface.html#s
ec:command_line_warnings in the Gradle documentation.

BUILD SUCCESSFUL in 3s
1 actionable task: 1 executed
[ec2-user@ip-172-31-31-40 Assignment1]$ gradle runFrac -Pnum=12 -Pdenom=5

> Task :runFrac
The fraction is: 12/5

[Incubating] Problems report is available at: file:///home/ec2-user/ser321-fall26C-scollenb/Assignment
1/build/reports/problems/problems-report.html

Deprecated Gradle features were used in this build, making it incompatible with Gradle 9.0.

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e from your own scripts or plugins.

For more on this, please refer to https://docs.gradle.org/8.13/userguide/command_line_interface.html#s
ec:command_line_warnings in the Gradle documentation.

BUILD SUCCESSFUL in 2s
2 actionable tasks: 2 executed
<-----> 0% WAITING
> IDLE
[ec2-user@ip-172-31-31-40 Assignment1]$ |
```

1-

Commands: gradle tasks

gradle runFrac -Pnum=12 -Pdenom=5

2.4 Screencast for second systems

<https://youtu.be/vXvsI9cGOJw>

Task 3

3.1

In powershell:

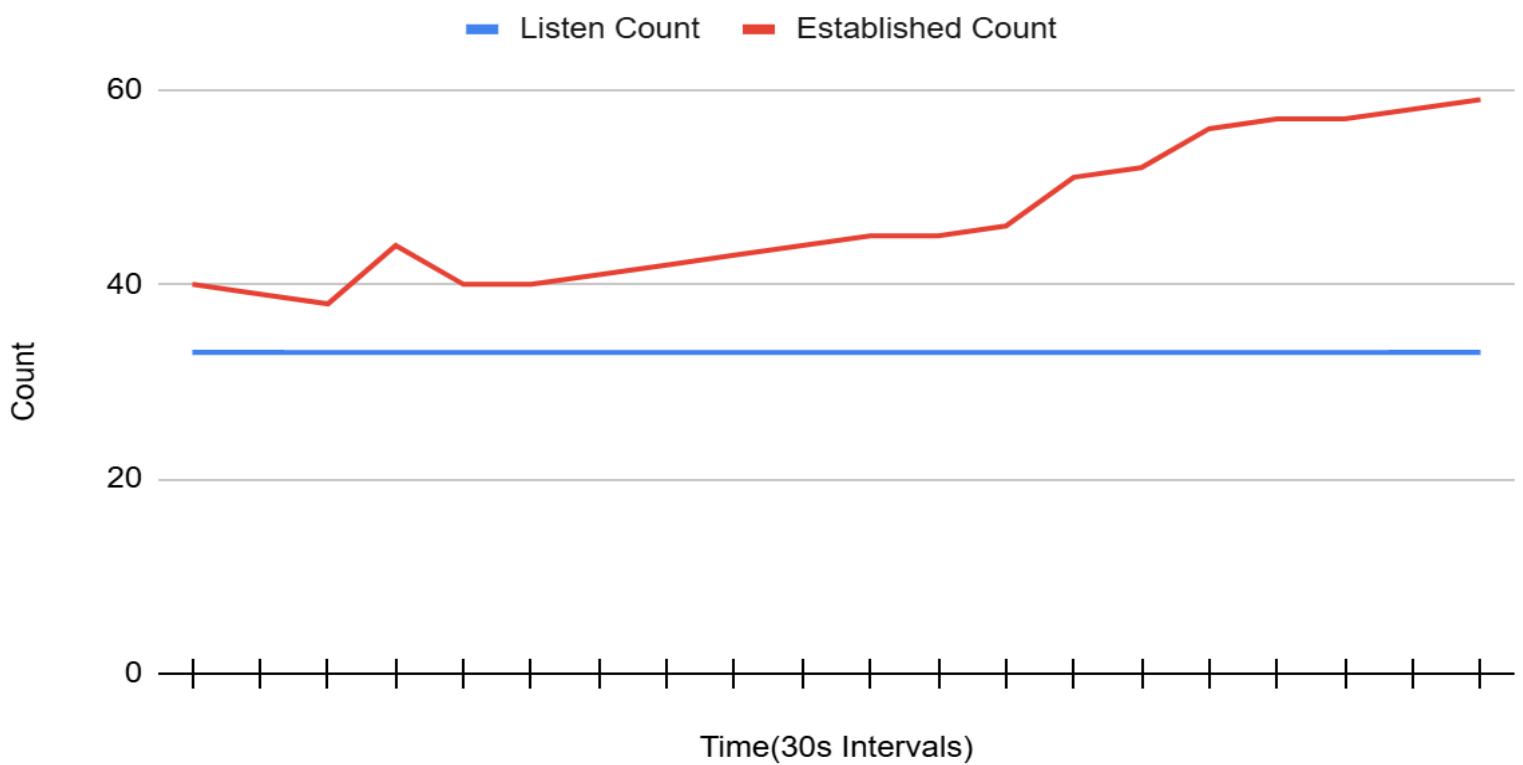
```
$first = Get-Date
```

```
$end = $first.AddMinutes(10)
```

```
$log = "netlog.txt"
```

```
while ((Get-Date) -lt $end) {  
    Get-Date | Out-File -Append $log  
    Get-NetTCPConnection |  
        Where-Object { $_.State -in @('Listen','Established') } |  
        Out-File -Append $log  
    "" | Out-File -Append $log  
    Start-Sleep -Seconds 30  
}
```

Established/ Listen connections



3.2

3.2.1

```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sgcol>nc -k -l 3333
'nc' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\sgcol>Ncat -k -l 3333
SER321
is absolutely amazing!!!
Networking is so fun
^C
C:\Users\sgcol>
```

```
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sgcol>Ncat 127.0.0.1 3333
SER321
is absolutely amazing!!!
Networking is so fun
^C
C:\Users\sgcol>
```

tcp.port == 3333							
No.	Time	Source	Destination	Protocol	Length	Info	
211	147.665276200	127.0.0.1	127.0.0.1	TCP	56	28308 → 3333	[SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SAC...
212	147.665319700	127.0.0.1	127.0.0.1	TCP	56	3333 → 28308	[SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495...
213	147.665334600	127.0.0.1	127.0.0.1	TCP	44	28308 → 3333	[ACK] Seq=1 Ack=1 Win=65280 Len=0
222	153.424687500	127.0.0.1	127.0.0.1	TCP	52	28308 → 3333	[PSH, ACK] Seq=1 Ack=1 Win=65280 Len=8
223	153.424705900	127.0.0.1	127.0.0.1	TCP	44	3333 → 28308	[ACK] Seq=1 Ack=9 Win=65280 Len=0
267	164.256491900	127.0.0.1	127.0.0.1	TCP	70	28308 → 3333	[PSH, ACK] Seq=9 Ack=1 Win=65280 Len=26
268	164.256514500	127.0.0.1	127.0.0.1	TCP	44	3333 → 28308	[ACK] Seq=1 Ack=35 Win=65280 Len=0
320	171.064119700	127.0.0.1	127.0.0.1	TCP	66	28308 → 3333	[PSH, ACK] Seq=35 Ack=1 Win=65280 Len=22
321	171.064136900	127.0.0.1	127.0.0.1	TCP	44	3333 → 28308	[ACK] Seq=1 Ack=57 Win=65280 Len=0
365	180.901067800	127.0.0.1	127.0.0.1	TCP	44	28308 → 3333	[FIN, ACK] Seq=57 Ack=1 Win=65280 Len=0
366	180.901146500	127.0.0.1	127.0.0.1	TCP	44	3333 → 28308	[ACK] Seq=1 Ack=58 Win=65280 Len=0
367	180.901271300	127.0.0.1	127.0.0.1	TCP	44	3333 → 28308	[FIN, ACK] Seq=1 Ack=58 Win=65280 Len=0
368	180.901354800	127.0.0.1	127.0.0.1	TCP	44	28308 → 3333	[ACK] Seq=58 Ack=2 Win=65280 Len=0

- Roughly how many packets were shown for the whole exchange (back and forth)?
Almost a dozen packets were sent back and forth
- Roughly how many total bytes were shown for the whole exchange (back and forth)?
Almost 600 byte, I believe were sent back and forth.
- How many bytes were the actual data (words) that you sent?
8 words were sent

3.2.2

```
Command Prompt
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sgcol>Ncat -k -l 3333
SER321
is absolutely amazing!!!
Networking is so fun
^C
C:\Users\sgcol>Ncat -k -l -u 3333
SER321
is absolutely amazing!!!
Networking is so fun.
^C
C:\Users\sgcol>

Command Prompt
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

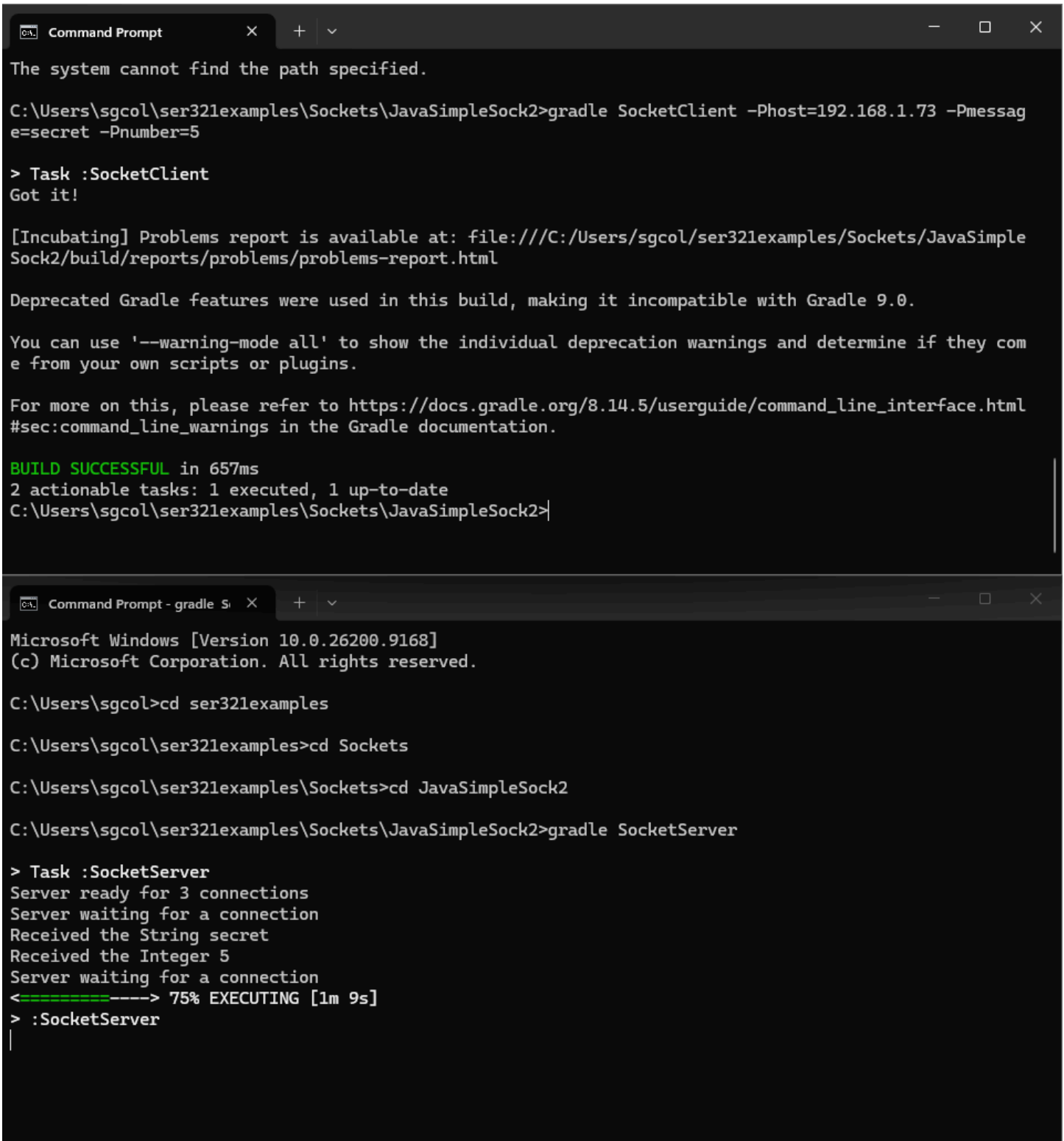
C:\Users\sgcol>Ncat 127.0.0.1 3333
SER321
is absolutely amazing!!!
Networking is so fun
^C
C:\Users\sgcol>Ncat -u 127.0.0.1 3333
SER321
is absolutely amazing!!!
Networking is so fun.
^C
C:\Users\sgcol>
```

udp.port==3333						
No.	Time	Source	Destination	Protocol	Length Info	
517	315.615527800	127.0.0.1	127.0.0.1	UDP	40	58652 → 3333 Len=8
571	327.529272900	127.0.0.1	127.0.0.1	UDP	58	58652 → 3333 Len=26
591	340.399565700	127.0.0.1	127.0.0.1	UDP	55	58652 → 3333 Len=23

- a) Roughly how many packets were shown for the whole exchange (back and forth)?
Three were shown
- b) Roughly how many total bytes were shown for the whole exchange (back and forth)?
About 150 were sent
- c) How many bytes were the actual data (words) that you sent?
8 words were sent

3.3

3.3.1 - <https://youtu.be/ELYMw9Jk214>



```
Command Prompt
The system cannot find the path specified.

C:\Users\sgcol\ser321examples\Sockets\JavaSimpleSock2>gradle SocketClient -Phost=192.168.1.73 -Pmessage=secret -Pnumber=5

> Task :SocketClient
Got it!

[Incubating] Problems report is available at: file:///C:/Users/sgcol/ser321examples/Sockets/JavaSimpleSock2/build/reports/problems/problems-report.html

Deprecated Gradle features were used in this build, making it incompatible with Gradle 9.0.

You can use '--warning-mode all' to show the individual deprecation warnings and determine if they come from your own scripts or plugins.

For more on this, please refer to https://docs.gradle.org/8.14.5/userguide/command_line_interface.html#sec:command_line_warnings in the Gradle documentation.

BUILD SUCCESSFUL in 657ms
2 actionable tasks: 1 executed, 1 up-to-date
C:\Users\sgcol\ser321examples\Sockets\JavaSimpleSock2>

Command Prompt - gradle S
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sgcol>cd ser321examples

C:\Users\sgcol\ser321examples>cd Sockets

C:\Users\sgcol\ser321examples\Sockets>cd JavaSimpleSock2

C:\Users\sgcol\ser321examples\Sockets\JavaSimpleSock2>gradle SocketServer

> Task :SocketServer
Server ready for 3 connections
Server waiting for a connection
Received the String secret
Received the Integer 5
Server waiting for a connection
<=====--> 75% EXECUTING [1m 9s]
> :SocketServer
|
```

tcp.port=8888						
No.	Time	Source	Destination	Protocol	Length	Info
60	1.918681300	192.168.1.73	192.168.1.73	TCP	56	59063 → 8888 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
61	1.918728000	192.168.1.73	192.168.1.73	TCP	56	8888 → 59063 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
62	1.918743300	192.168.1.73	192.168.1.73	TCP	44	59063 → 8888 [ACK] Seq=1 Ack=1 Win=65280 Len=0
63	1.920043300	192.168.1.73	192.168.1.73	TCP	48	59063 → 8888 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=4
64	1.920060700	192.168.1.73	192.168.1.73	TCP	44	8888 → 59063 [ACK] Seq=1 Ack=5 Win=65280 Len=0
65	1.924296400	192.168.1.73	192.168.1.73	TCP	53	59063 → 8888 [PSH, ACK] Seq=5 Ack=1 Win=65280 Len=9
66	1.924312300	192.168.1.73	192.168.1.73	TCP	44	8888 → 59063 [ACK] Seq=1 Ack=14 Win=65280 Len=0
67	1.924903700	192.168.1.73	192.168.1.73	TCP	84	59063 → 8888 [PSH, ACK] Seq=14 Ack=1 Win=65280 Len=40
68	1.924916500	192.168.1.73	192.168.1.73	TCP	44	8888 → 59063 [ACK] Seq=1 Ack=54 Win=65280 Len=0
69	1.924947500	192.168.1.73	192.168.1.73	TCP	75	59063 → 8888 [PSH, ACK] Seq=54 Ack=1 Win=65280 Len=31
70	1.924950900	192.168.1.73	192.168.1.73	TCP	44	8888 → 59063 [ACK] Seq=1 Ack=85 Win=65280 Len=0
71	1.925085100	192.168.1.73	192.168.1.73	TCP	46	59063 → 8888 [PSH, ACK] Seq=85 Ack=1 Win=65280 Len=2
72	1.925090200	192.168.1.73	192.168.1.73	TCP	44	8888 → 59063 [ACK] Seq=1 Ack=87 Win=65280 Len=0
73	1.925101700	192.168.1.73	192.168.1.73	TCP	48	59063 → 8888 [PSH, ACK] Seq=87 Ack=1 Win=65280 Len=4
74	1.925104800	192.168.1.73	192.168.1.73	TCP	44	8888 → 59063 [ACK] Seq=1 Ack=91 Win=65280 Len=0
75	1.925300800	192.168.1.73	192.168.1.73	TCP	48	8888 → 59063 [PSH, ACK] Seq=1 Ack=91 Win=65280 Len=4
76	1.925309200	192.168.1.73	192.168.1.73	TCP	44	59063 → 8888 [ACK] Seq=91 Ack=5 Win=65280 Len=0
77	1.925336700	192.168.1.73	192.168.1.73	TCP	54	8888 → 59063 [PSH, ACK] Seq=5 Ack=91 Win=65280 Len=10
78	1.925340300	192.168.1.73	192.168.1.73	TCP	44	59063 → 8888 [ACK] Seq=91 Ack=15 Win=65280 Len=0
85	1.930639200	192.168.1.73	192.168.1.73	TCP	44	59063 → 8888 [FIN, ACK] Seq=91 Ack=15 Win=65280 Len=0
86	1.930649400	192.168.1.73	192.168.1.73	TCP	44	8888 → 59063 [ACK] Seq=15 Ack=92 Win=65280 Len=0

3.3.2 - <https://youtu.be/TwvzsOG6sOw>

```

BUILD SUCCESSFUL in 751ms
2 actionable tasks: 1 executed, 1 up-to-date
C:\Users\sgcol\ser321examples\Sockets\JavaSimpleSock2>gradle SocketClient -Phost=54.221.48.247 -Pmessage=Duh -Pnumber=25

> Task :SocketClient
Got it!

[Incubating] Problems report is available at: file:///C:/Users/sgcol/ser321examples/Sockets/JavaSimpleSock2/build/reports/problems/problems-report.html

Deprecated Gradle features were used in this build, making it incompatible with Gradle 9.0.

You can use '--warning-mode all' to show the individual deprecation warnings and determine if they come from your own scripts or plugins.

For more on this, please refer to https://docs.gradle.org/8.14.5/userguide/command_line_interface.html#sec:command_line_warnings in the Gradle documentation.

BUILD SUCCESSFUL in 821ms
2 actionable tasks: 1 executed, 1 up-to-date
C:\Users\sgcol\ser321examples\Sockets\JavaSimpleSock2>

```

```

BUILD SUCCESSFUL in 3m 6s
2 actionable tasks: 1 executed, 1 up-to-date
[ec2-user@ip-172-31-31-40 JavaSimpleSock2]$ gradle SocketServer

> Task :SocketServer
Server ready for 3 connections
Server waiting for a connection
Received the String Duh
Received the Integer 25
Server waiting for a connection
<=====----> 75% EXECUTING [59s]
> :SocketServer

```

tcp.port==8888						
No.	Time	Source	Destination	Protocol	Length	Info
300	17.183218500	54.221.48.247	192.168.1.73	TCP	66	8888 → 55592 [SYN, ACK] Seq=0 Ack=1 Win=62727 Len=0 MSS=1460 SACK_PERM WS=256
309	17.366100400	54.221.48.247	192.168.1.73	TCP	64	8888 → 55592 [PSH, ACK] Seq=5 Ack=88 Win=62976 Len=10
306	17.279423600	54.221.48.247	192.168.1.73	TCP	58	8888 → 55592 [PSH, ACK] Seq=1 Ack=88 Win=62976 Len=4
312	17.448144500	54.221.48.247	192.168.1.73	TCP	54	8888 → 55592 [ACK] Seq=15 Ack=89 Win=62976 Len=0
305	17.260805100	54.221.48.247	192.168.1.73	TCP	54	8888 → 55592 [ACK] Seq=1 Ack=88 Win=62976 Len=0
303	17.222413200	54.221.48.247	192.168.1.73	TCP	54	8888 → 55592 [ACK] Seq=1 Ack=5 Win=62976 Len=0
299	17.146340200	192.168.1.73	54.221.48.247	TCP	66	55592 → 8888 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
304	17.222442800	192.168.1.73	54.221.48.247	TCP	137	55592 → 8888 [PSH, ACK] Seq=5 Ack=1 Win=65280 Len=83
302	17.185184000	192.168.1.73	54.221.48.247	TCP	58	55592 → 8888 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=4
310	17.367019700	192.168.1.73	54.221.48.247	TCP	54	55592 → 8888 [FIN, ACK] Seq=88 Ack=15 Win=65280 Len=0
307	17.329637100	192.168.1.73	54.221.48.247	TCP	54	55592 → 8888 [ACK] Seq=88 Ack=5 Win=65280 Len=0
301	17.183298000	192.168.1.73	54.221.48.247	TCP	54	55592 → 8888 [ACK] Seq=1 Ack=1 Win=65280 Len=0

So... I misread the task and ran the AWS/Command prompt one first. Neither operated differently to me, in the actual gradle commands or in wireshark. Both sets operated identically, from a purely implementation stance. The only code I had to change was the host ID to connect to the AWS and command prompt individually. The wireshark output shows a different story, as the command prompt implementation took almost twice the amount of packets to implement.